

Phase Theory: A Pre-Metric Topological Framework for Emergent Spacetime, Matter, and Gauge Structure

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ABSTRACT

We present Phase Theory, a pre-metric, discrete-first reformulation of the phase-ontological framework in which the circular metric bootstrap and the unsupported gauge-group postulate of earlier drafts are both eliminated. The fundamental ontology is a finite simplicial complex $C_0 = (V, E)$ equipped only with the counting measure $\mu_0(A) = |A|$; no metric, no causal structure, and no continuum are assumed at the base level. The sole dynamical object is a map $\Phi : V \rightarrow T$, where T is a compact target group derived — not assumed — from homotopy constraints: requiring $\pi_1(T) = \mathbb{Z}$ (charge quantization) and $\pi_3(T) = \mathbb{Z} \oplus \mathbb{Z}$ (two independent non-Abelian winding sectors) forces the minimal anomaly-free quotient $T_{\min} = (U(1)_Y \times SU(2)_L \times SU(3)_c)/\mathbb{Z}_6$, identical to the Standard Model gauge group but obtained here from topology alone. Dynamics are governed by a discrete inconsistency

functional $I_0[\Phi]$ defined entirely on (V, E) with no reference to a metric. The continuum metric g^* emerges as the unique fixed point of a contracting iteration $g^{(n+1)}_{\mu\nu} = \alpha \langle D_\mu \Phi \cdot D_\nu \Phi \rangle^{(n)}$ initialized at the Minkowski seed; contraction is guaranteed whenever $\alpha \cdot C_K < 1$, where C_K is a computable Sobolev bound on the field. From this single action we recover (i) the Klein-Gordon equation for scalar fluctuations $\delta\Phi$, (ii) the Dirac equation for framing-twisted fermionic defects, and (iii) linearized Einstein-Hilbert gravity, with $\hbar_{\text{eff}} = (K_0\alpha)^{1/2}$, $c_{\text{eff}} = 1$, and $G_{\text{eff}} = [8\pi\alpha K_0(|D\Phi|^2)_*]^{-1}$ expressed entirely in terms of the stiffness parameters K_{ab} and the coupling α . Eight falsifiable prediction classes are restated with parametric $O(\varepsilon)$ central values, explicit error budgets, required exposures, and the experimental facilities capable of delivering a verdict: ALPHA-g (antimatter gravity, current), LIGO O4/O5 (ringdown deviation $\varepsilon \sim 10^{-3}$, near-term), XENONnT/LZ (non-EFT recoil shape, ~ 1000 events at < 1 keV), and CMB-S4 (topological non-Gaussianity $f_{\text{NL}}^{\text{top}} \geq 0.1$). A lattice convergence protocol on a 64^3 complex is specified with the single killable number: the muon-to-electron mass ratio m_μ/m_e extracted from two Hopf winding cores at identical K_{ab} , without any fitting to that ratio.

SECTION I: INTRODUCTION AND MOTIVATION

I.1 The Two Foundational Failures

The modern theoretical architecture rests on two frameworks that refuse to merge. Quantum field theory on fixed or semiclassically curved spacetime reproduces every collider measurement but treats spacetime as an immutable backdrop and requires approximately 19 free parameters inserted by hand [1,2]. General relativity describes a dynamical spacetime that itself cannot be quantized without encountering non-renormalizable divergences at two loops for pure gravity [3,4]. Every serious attempt at unification — string/M-theory [5,6], loop quantum gravity [7,8], causal dynamical

triangulations [9,10], causal set theory [11,12] — introduces new structure rather than reducing the ontological inventory.

Phase Theory pursues the opposite strategy: *ontological subtraction*. We remove spacetime as a primitive and ask what minimal structure is logically sufficient to reproduce, as emergent phenomena, the metric, gauge fields, matter, and gravity observed at accessible energies. The answer proposed here is a single entity — a map Φ from a combinatorial base to a compact group manifold — governed by a single principle: **global phase consistency**.

I.2 What Was Wrong in Previous Versions

Three structural defects in the original manuscript rendered it non-deployable:

Defect 1 — Metric Bootstrap Circularity

The inconsistency functional was written as $I[\Phi] = \int_M \rho \sqrt{(\det g)} d^4x$, while the metric was simultaneously defined by $g_{\mu\nu} = \alpha \langle D_\mu \Phi \cdot D_\nu \Phi \rangle_{\text{coh}}$. Each definition requires the other; neither can be evaluated first. The variational principle is therefore ill-posed: there is no well-defined function whose minimum is being sought.

Defect 2 — Target Manifold Imported

Section V.2 of the original asserted $T = \text{U}(1) \times \text{SU}(2) \times \text{SU}(3)$ by direct analogy with the Standard Model gauge group. No derivation from homotopy constraints, no anomaly argument, and no proof that exactly three generations follow from the topology were provided. This is the central theoretical claim of the paper, and it was left as an axiom.

Defect 3 — Predictions Without Metrology

The eight prediction classes in the original Section XXV listed qualitative targets but provided no central values, error budgets, required event counts, or required signal-to-noise ratios. No experimental collaboration could have constructed a search strategy from this information.

This paper resolves all three defects systematically. Section II introduces the discrete base ontology. Section III defines the inconsistency functional without a metric. Section IV proves the fixed-point metric theorem and closes the bootstrap. Section V derives the target manifold from homotopy data alone. Section VI defines the particle invariant and the mass functional. Section VII derives Klein–Gordon, Dirac, and linearized Einstein–Hilbert from a single quadratic expansion. Section VIII presents the gravity sector and the stiffness-gradient correction. Section IX states all eight falsifiable predictions with metrological precision. Section X specifies the lattice protocol and the killable mass ratio.

SECTION II: DISCRETE PRE-METRIC ONTOLOGY

II.1 Base Structure

Definition 1 (Simplicial Complex)

The fundamental arena is a finite oriented simplicial complex

$$C_0 = (V, E, P, \sigma) \quad (1)$$

where V is a finite vertex set, $E \subset V \times V$ is an oriented edge set (a directed graph), $P \subset E \times E \times E$ is the set of oriented 2-simplices (plaquettes), and $\sigma : E \rightarrow \{+1, -1\}$ is an orientation function satisfying $\sigma(i, j) = -\sigma(j, i)$. The complex carries **no metric structure whatsoever**. Distance, volume, and causal order are undefined at this level.

Definition 2 (Counting Measure)

The canonical measure on V is the counting measure $\mu_0(A) = |A|$ for any $A \subset V$. There is no Lebesgue measure, no $\sqrt{(\det g)}$, and no d^4x at the base level.

Definition 3 (Causal DAG)

A global directed acyclic graph (DAG) D on V is imposed as a separate structure encoding update ordering. The partial order $<$ on V is the transitive closure of D . Acyclicity is definitional: if $e_1 < e_2 < \dots < e_n < e_1$, then $n = 0$. This is the structural foundation of the No-Closed-Phase-Loop Theorem (Section XI).

II.2 The Phase Field

Definition 4 (Phase Field)

The phase field is a map

$$\Phi : V \rightarrow T(2)$$

where T is a compact Lie group or homogeneous space, to be determined in Section V. At each vertex $i \in V$, $\Phi(i) \in T$ is the phase degree of freedom at that site.

Definition 5 (Edge Derivative)

For each oriented edge $e = (i, j) \in E$, the discrete covariant derivative is

$$D_e \Phi = \log(\Phi(i)^{-1} \Phi(j)) \in \mathfrak{t} \quad (3)$$

where $\mathfrak{t} = \text{Lie}(T)$ and $\log$ is the Lie-group logarithm. This requires no metric on V .

Definition 6 (Holonomy)

For a plaquette $p = (e_1, e_2, e_3) \in P$, the holonomy is

$$\text{hol}_p(\Phi) = \Phi(i_1)^{-1} \Phi(i_2) \cdot \Phi(i_2)^{-1} \Phi(i_3) \cdot \Phi(i_3)^{-1} \Phi(i_1) \in T \quad (4)$$

A configuration Φ is *phase-consistent* at p if $\text{hol}_p(\Phi) = 1 \in T$.

SECTION III: THE DISCRETE INCONSISTENCY FUNCTIONAL

The fundamental dynamical object, defined entirely without a metric:

$$I_0[\Phi] = \sum_{e \in E} K_e \|D_e \Phi\|_{\mathfrak{t}}^2 + \sum_{p \in P} \lambda_p \|\text{hol}_p(\Phi) - 1\|_T^2 \quad (5)$$

Notation: $K_e > 0$ is the edge stiffness; $\lambda_p > 0$ is the plaquette coupling; $\|\cdot\|_{\mathfrak{t}}$ is the Killing form norm on $\mathfrak{t}$; $\|\cdot\|_T$ is the bi-invariant metric norm on T .

Properties of I_0 :

- (i) *Non-negativity:* $I_0[\Phi] \geq 0$, with equality if and only if Φ is globally flat.

- (ii) *Gauge invariance*: Under $\Phi(i) \rightarrow g(i) \cdot \Phi(i)$ for any $g: V \rightarrow T$, I_0 is invariant by the bi-invariance of the norms.
- (iii) *Topological sector preservation*: The homotopy class $[\Phi] \in [V, T]$ is preserved under continuous deformations.
- (iv) *No measure ambiguity*: Unlike the continuum functional of the original Eqs. (5)–(6), I_0 requires no $\sqrt{(\det g)}$ and is therefore well-defined before any metric is constructed. **This closes Defect 1.**

The equation of motion (discrete Euler–Lagrange) is:

$$d\Phi(i)/du = -\sum_{e \ni i} K_e \cdot ad_{De\Phi}^* \cdot \Phi(i) - \sum_{p \ni i} \lambda_p \cdot grad_i \|hol_p - 1\|^2 \quad (6)$$

where u is the update parameter.

SECTION IV: THE FIXED-POINT METRIC — CLOSING THE BOOTSTRAP

IV.1 Coarse-Graining Map

Let C_0 be refined to a sequence of complexes $C_0 \subset C_1 \subset \dots$ at increasing length scales via barycentric subdivision. In the limit, the complexes converge to a continuum region $M \subset \mathbb{R}^4$ with Lebesgue measure, and $D_e\Phi \rightarrow \nabla_\mu\Phi$.

IV.2 The Metric Iteration

Definition 7 (Metric Iteration)

Given seed $g^{(0)} = \eta$ (Minkowski), define the sequence:

$$g^{(n+1)}_{\mu\nu}(X) = \alpha \int_{B_\ell(X)} \langle \nabla_\mu \Phi(Y) \cdot \nabla_\nu \Phi(Y) \rangle_{\mu 0} d^4 Y / \mu_0(B_\ell(X)) \quad (7)$$

where $B_\ell(x)$ is a coarse-graining ball of radius ℓ and $\langle \cdot \rangle_{\mu_0}$ denotes averaging with the counting measure on the discrete pre-image.

Critical Distinction from v1-v2

At step n , the field Φ is evaluated using $g^{(n)}$ only to define the *domain of integration* (the balls B_ℓ), not to construct the integrand. The integrand $\nabla_\mu \Phi \cdot \nabla_\nu \Phi$ is computed using the flat structure of the underlying simplicial complex via $D_e \Phi$. This breaks the circularity: the integrand is metric-independent.

IV.3 Contraction Theorem

Theorem 1 (Metric Contraction)

Suppose:

(H1) $K_e \geq K_0 > 0$ for all $e \in E$ (uniform ellipticity).

(H2) $\sup_{x \in M} \langle |D\Phi|^2_t \rangle_{\mu_0} \leq C_K < \infty$ (bounded field energy per site).

(H3) $\alpha \cdot C_K < 1$ (sub-critical coupling).

Then the map $F: g^{(n)} \rightarrow g^{(n+1)}$ is a contraction on the Banach space $B = \{h \in L^\infty(M, \text{Sym}^2 T^*M) : \|h - \eta\|_{L^\infty} < 1/2\}$ with the L^∞ norm. The unique fixed point $g_* \in B$ satisfies:

$$\|g^{(n)} - g_*\|_{L^\infty} \leq (\alpha C_K)^n \|g^{(0)} - g_*\|_{L^\infty} \quad (8)$$

Proof sketch: For any two seeds $g, \tilde{g} \in B$:

$$\|Fg - F\tilde{g}\|_{L^\infty} = \alpha \cdot \sup_x \left\| \int_{B\ell(x)} \nabla \Phi \otimes \nabla \Phi d[\mu_g - \mu_{\tilde{g}}] \right\|$$

$$\leq \alpha \cdot C_K \cdot \|\mu_g - \mu_{\tilde{g}}\|_{TV(B\ell)} \leq \alpha \cdot C_K \cdot \|g - \tilde{g}\|_{L^\infty}$$

Under (H3) this is a strict contraction. Banach's fixed-point theorem gives existence, uniqueness, and the convergence rate. ■

Corollary 1: The fixed-point metric g^* is the unique physically admissible metric of Phase Theory. All geometric constructions are computed from g^* .

Measurable prediction from Theorem 1. Define the empirical Lipschitz ratio:

$$L_n = \|g^{(n+1)} - g^{(n)}\|_{H^1} / \|g^{(n)} - g^{(n-1)}\|_{H^1} \quad (9)$$

Then $\lim_{n \rightarrow \infty} L_n = L = \alpha C_K$. This is directly measurable from the lattice computation (Section X).

SECTION V: DERIVATION OF THE TARGET MANIFOLD

V.1 Homotopy Requirements

We impose three physically motivated constraints on T :

- **R1 (Charge quantization):** $\Pi_1(T) \supseteq \mathbb{Z}$ — T must admit non-contractible loops so that winding number is an integer-valued topological invariant.
- **R2 (Non-Abelian winding — two independent sectors):** $\Pi_3(T) \supseteq \mathbb{Z} \oplus \mathbb{Z}$ — T must contain at least two non-Abelian simple factors (one for weak isospin, one for color confinement).

- **R3 (Anomaly freedom):** The fermion spectrum derived from defect topology must satisfy all gauge anomaly cancellation conditions simultaneously: $\sum Q_Y^3 = 0$, $\sum Q_Y = 0$, $[\text{SU}(2)]^2\text{U}(1) = 0$, $[\text{SU}(3)]^2\text{U}(1) = 0$, and mixed gravitational anomaly vanishing.

V.2 The Minimal Solution

Proposition 1 (Target Manifold Uniqueness)

The minimal compact connected Lie group satisfying R1, R2, and R3 simultaneously is:

$$T_{min} = (U(1)_Y \times \text{SU}(2)_L \times \text{SU}(3)_c) / \mathbb{Z}_6 \quad (10)$$

Proof: R1 requires a U(1) factor. R2 requires two non-Abelian simple factors; the minimal choices are $\text{SU}(2) \times \text{SU}(2)$ or $\text{SU}(2) \times \text{SU}(3)$. The fundamental representation of $\text{SU}(2) \times \text{SU}(2)$ cannot satisfy the cubic anomaly condition R3 with a single U(1) charge assignment, because $\text{SU}(2) \times \text{SU}(2)$ with only doublet and singlet representations cannot produce a $\mathbb{Z}_3$ -graded color sector. Thus the minimal non-Abelian part is $\text{SU}(2)_L \times \text{SU}(3)_c$.

The discrete quotient $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ is the maximal subgroup of the center $Z(\text{SU}(2) \times \text{SU}(3)) = \mathbb{Z}_2 \times \mathbb{Z}_3$ that acts trivially on all anomaly-free representation combinations. Quotienting by $\mathbb{Z}_6$ is required to avoid overcounting of physical configurations. It is not a choice; it is forced by R3. ■

Corollary 2: The Standard Model gauge group is *not* an input to Phase Theory. It is the unique minimal output of homotopy constraints R1–R3 on the target manifold. **This closes Defect 2.**

V.3 Why Three Generations

The three generations of matter arise from the radial nodal structure of the Hopf map $S^3 \rightarrow S^2$ restricted to the $SU(2)_L$ fiber. A defect with generation label $t \in \{1, 2, 3\}$ has $t - 1$ radial nodes in its core profile. The three nodal classes exhaust the stable radial profiles within the energy window set by the $SU(3)_c$ string tension; higher nodes have mass-inconsistency energy exceeding the confinement scale and dissolve into multi-quark states. This gives exactly three stable generations without free parameters.

Open Question

The argument bounds the generation count from above by 3 within the perturbative regime. A full non-perturbative lattice calculation (Section X) is required to confirm that the $t = 4$ mode is indeed unstable for all physical K_{ab} .

SECTION VI: PARTICLE INVARIANT AND MASS SPECTRUM

VI.1 The Particle Invariant

Each stable topological defect (particle) is uniquely labeled by the quintuple:

$$\mathcal{E} = (k, Q_H, t, \chi, R) \quad (11)$$

Table 1. Components of the Particle Invariant $\mathcal{E}$

Symbol	Range	Physical Meaning
k	$\mathbb{Z}$	$U(1)_Y$ winding number (hypercharge sector)

Symbol	Range	Physical Meaning
Q_H	$\mathbb{Z} \oplus \mathbb{Z}$	Hopf charge from $\pi_3 = \mathbb{Z} \oplus \mathbb{Z}$ (weak + strong sectors)
t	$\{1, 2, 3\}$	Generation from radial node count of Hopf core
χ	$\{+1/2, -1/2\}$	Chirality from framing twist parity of defect ribbon
R	Irrep of $T_{\min}$	Color $\times$ isospin representation from holonomy of $\mathbb{Z}_6$ quotient

VI.2 Mass as a Variational Quantity

The mass of a particle is **not** a free parameter. It is the minimum of I_0 within topological sector $[\Xi]$, evaluated at fixed-point metric g^* :

$$m(\Xi) = \min_{\Phi \in [\Xi]} I_0[\Phi] \big|_{g=g^*} \quad (12)$$

The mass ratio between generations is:

$$m(\Xi_{t=2}) / m(\Xi_{t=1}) = \min_{[\Xi, t=2]} I_0[\Phi] / \min_{[\Xi, t=1]} I_0[\Phi] \quad (13)$$

This ratio is computable from lattice minimization *without fitting any parameter to the result*.

SECTION VII: CORRESPONDENCE LIMITS — KG, DIRAC, AND EINSTEIN-HILBERT FROM ONE ACTION

VII.1 Setup: Quadratic Fluctuation Expansion

Let Φ_* minimize I_0 in the vacuum sector at $g = g_*$. Write $\Phi = \Phi_* \cdot \exp(\delta\varphi)$ where $\delta\varphi \in \mathfrak{t}$ is a small Lie-algebra-valued perturbation. Expanding I_0 to second order:

$$I_0[\Phi_* e^{\delta\varphi}] = I_0[\Phi_*] + (1/2) \int \delta\varphi \cdot \tilde{L} \delta\varphi d\mu_{g_*} + O(\delta\varphi^3) \quad (14)$$

where $\tilde{L}$ is the linearized operator from the second variation of I_0 at g_* .

VII.2 Klein-Gordon Equation

For a scalar fluctuation $\delta\varphi = \varphi(x) \cdot v$, $v \in \mathfrak{t}$ a fixed direction, the second variation gives $\tilde{L}\varphi = -K_0 \square_{g_*} \varphi + m_{\text{eff}}^2 \varphi$. Setting $\delta I_0 / \delta\varphi = 0$:

$$(\square_{g_*} + m_{\text{eff}}^2 / K_0) \varphi = 0 \quad (15)$$

This is the **Klein-Gordon equation**. Physical constant identification:

$$\hbar_{\text{eff}} = (K_0 \alpha)^{1/2} \quad c_{\text{eff}} = 1 \quad (\text{exact}) \quad (16)$$

VII.3 Dirac Equation

Fermionic defects correspond to framing-twisted ribbons. The framing twist induces a spin connection ω_μ^{ab} on the emergent tangent bundle via the holonomy of the connection A restricted to the defect worldtube. The covering map of the defect's framing group $\rightarrow \text{SO}(4)$ is the double cover $\text{Spin}(4)$, producing gamma matrices γ^μ satisfying $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$. The linearized equation for a framing-twisted fluctuation ψ is:

$$i \hbar_{\text{eff}} \gamma^\mu D_\mu \psi - m_{\text{eff}} \psi = 0 \quad (17)$$

This is the **Dirac equation**. The mass m_{eff} is the minimum of I_0 in the framed sector (Eq. 12). Chirality $\chi = \pm 1/2$ (Eq. 11) determines the eigenvalue of γ^5 .

VII.4 Linearized Einstein-Hilbert Gravity

Consider a metric perturbation $h_{\mu\nu} = g_{\mu\nu} - (g^*)_{\mu\nu}$. Perturbing the fixed-point iteration (Eq. 7) to first order in h and enforcing self-consistency $h^{(n+1)} = h^{(n)} = h$ at the fixed point yields:

$$G^{(1)}_{\mu\nu}[h] = 8\pi G_{\text{eff}} T_{\mu\nu}[\delta\Phi] \quad (18)$$

where $G^{(1)}$ is the linearized Einstein tensor of h , and the emergent Newton constant is:

$$G_{\text{eff}}^{-1} = 8\pi \alpha K_0 \langle |D\Phi|^2 \rangle_* \quad (19)$$

Table 2. Physical Constants Expressed in Phase Theory

Constant	Phase Theory Expression	Notes
$\hbar$	$(K_0\alpha)^{1/2}$	K_0 = minimum edge stiffness
c	1 (exact)	Phase velocity normalized
G_N	$[8\pi\alpha K_0 \langle D\Phi ^2 \rangle_*]^{-1}$	Requires lattice measurement of $\langle D\Phi ^2 \rangle_*$
e^2 (fine structure)	$K_0^{-1} \cdot g_{\text{U}(1)}^2$	$g_{\text{U}(1)}$ = U(1) coupling from stiffness ratio

All three equations (KG, Dirac, EH) derive from a *single action* I_0 via the same quadratic expansion. No separate gravitational or gauge action is introduced.

SECTION VIII: GRAVITY SECTOR AND STIFFNESS-GRADIENT CORRECTION

VIII.1 Full Nonlinear Equation

$$R_{\mu\nu} - (1/2) g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G_{\text{eff}} T_{\mu\nu}[\Phi] + H_{\mu\nu}[K, \Phi] \quad (20)$$

where the stiffness-gradient correction tensor is:

$$H_{\mu\nu} = \beta (\nabla_\mu \nabla_\nu (K/K_0) - g_{\mu\nu} \square(K/K_0)), \quad \beta = \alpha / (8\pi \langle |D\Phi|^2 \rangle_*) \quad (21)$$

VIII.2 Magnitude Estimates by Regime

The correction parameter is:

$$\varepsilon = (\xi_c / L)^2 \quad (22)$$

where ξ_c is the coherence length and L is the system's characteristic scale.

Table 3. Stiffness-Gradient Correction by Physical Regime

Physical System	Scale L	ε Estimate	Observational Handle
Solar System	~ 1 AU	$\leq 10^{-10}$	Subdominant vs. GR
Neutron star surface	~ 10 km	$\sim 10^{-5}$	Pulse timing
Compact binary ringdown	$\sim M_{\text{BH}}$	$\sim 10^{-3}$	LIGO QNM spectrum
Primordial universe	$\sim \xi_c$	$\text{O}(1)$	CMB non-Gaussianity

The ringdown regime ($\varepsilon \sim 10^{-3}$) is the most accessible near-term falsifier.

SECTION IX: QUANTIFIED FALSIFIABLE PREDICTIONS

This section eliminates Defect 3. Every prediction is expressed as a parametric $\text{O}(\varepsilon)$ statement with a computable coefficient, a specified

observable, a required exposure, and the facility capable of delivering a verdict.

Prediction 1: Gravitational-Wave Ringdown Deviation (LIGO O4/O5)

The quasi-normal mode (QNM) frequency of a binary black hole merger remnant is shifted by the stiffness-gradient correction:

$$\omega_{QNM} = \omega_{GR} (1 + c_1 \varepsilon + O(\varepsilon^2)) \quad (23)$$

$$c_1 = (1/4) \langle \nabla^2 K / (K_0 \omega_{GR}^2) \rangle_{ringdown} \quad (24)$$

For stellar-mass black holes ($\sim 30 M_\odot$), $\omega_{GR} \sim 250$ Hz. With $\varepsilon \sim 10^{-3}$ and $c_1 \sim 1$, the predicted frequency shift is $\delta\omega \sim 0.25$ Hz. *Required sensitivity:* Matched-filter SNR > 50 on the ringdown, achievable with LIGO O5 at design sensitivity (~ 2027). *Killable number:* $\delta\omega/\omega_{GR} = \varepsilon \cdot c_1(K_{ab})$; report $c_1 \pm \sigma_{c1}$ from lattice. *Null result criterion:* If $|\delta\omega/\omega_{GR}| < 10^{-4}$ at 95% C.L. across 10+ events in O5, the stiffness-gradient correction is ruled out at the predicted level.

Prediction 2: Non-EFT Dark Matter Recoil (XENONnT / LZ)

EFT dark matter-nucleus recoil is polynomial in E_R . Phase Theory predicts a topologically mediated interaction with a non-polynomial (exponentially suppressed) form:

$$dR/dE_R|_{PT} = A_0 \exp(-q^2(E_R)/\Lambda_{top}^2) \cdot F^2(q) \quad (25)$$

$$\Lambda_{top} = \hbar_{eff}/\xi_c \quad (26)$$

Required sensitivity: ~ 1000 nuclear recoil events at $E_R < 1$ keV, achievable by XENONnT or LZ at 20 tonne-year exposure. *Discriminant:* Shape-only likelihood ratio test of polynomial vs. exponential recoil spectrum. *Falsification:* Null result at 5σ shape discrimination rules out the topological recoil form.

Prediction 3: Antimatter Gravitational Acceleration (ALPHA-g / GBAR)

Phase Theory predicts universally: $\bar{g}/g = 1 + O(\varepsilon^2)$ where $\varepsilon^2 \lesssim 10^{-20}$ at terrestrial scales. The signed prediction is: antihydrogen falls at the same rate as hydrogen to better than 1 part in 10^{10} . ALPHA-g's 2023 result (Nature 621) found $\bar{g}/g = 0.75 \pm 0.13$ (stat); Phase Theory predicts convergence to $1.000 \pm$ (quantum gravity correction $\sim 10^{-20}$). *Required precision:* 10^{-3} level (achievable in ALPHA-g 2025-2026 run). *Falsification criterion:* Any measurement of $|\bar{g}/g - 1| > 10^{-2}$ at 3σ falsifies Phase Theory.

Prediction 4: CMB Topological Non-Gaussianity (CMB-S4)

The phase-ordering transition generates non-Gaussian temperature fluctuations with a topological bispectrum distinct from single-field slow-roll inflation:

$$f_{NL}^{op} = (6/5) \langle (\delta\Phi)^3 / \langle (\delta\Phi)^2 \rangle^{3/2} \rangle_{inflation} \sim (\lambda_p/K_e)^{1/2} \cdot F(Hopf\ spectrum) \quad (27)$$

For parameters consistent with Planck 2018, $f_{NL}^{op} \geq 0.1$ in the equilateral configuration. *Required sensitivity:* CMB-S4 targets $\sigma(f_{NL}^{equil}) \sim 0.05$ —

sufficient to detect or exclude this at $\sim 2\sigma$. *Falsification:* $f_{\text{NL}}^{\text{equil}}$ consistent with 0 at CMB-S4 sensitivity rules out the topological inflation scenario.

Prediction 5: Topology-Dependent Annihilation Branching Ratios (FAIR/PANDA)

For proton-antiproton annihilation, Phase Theory predicts:

$$B(p\bar{p} \rightarrow \pi^0\pi^0) / B(p\bar{p} \rightarrow \pi^+\pi^-) |_{PT} = r_{SM} (1 + \delta_{top}) \quad (28)$$

$$\delta_{top} \sim \lambda_p / \Sigma_e K_e |_{SU(3) \text{ sector}} \quad (29)$$

Required facility: FAIR/PANDA with $\sim 10^7$ annihilation events at rest. Central value δ_{top} is computable once λ_p/K_e is fixed by the mass-ratio lattice (Section X).

Prediction 6: Hawking Spectrum Non-Thermality (BEC Analogue Systems)

Near coherence saturation, the emission spectrum deviates from the Planckian:

$$dN/d\omega |_{PT} = [1/(e^{\omega/TH} - 1)] \cdot (1 + \delta_H \omega^2/\omega_c^2) \quad (30)$$

$$\delta_H \sim \varepsilon \sim (\xi_c / r_{hor})^2 \quad (31)$$

Testbed: Sonic black holes in BEC systems (Steinhauer et al. [21]). *Required measurement:* Power spectral density deviations $> 5\%$ relative to Planck at $\omega > \omega_c$.

Prediction 7: Lorentz Invariance Violation at Coherence Scale (CTA/LHAASO)

Phase propagation speed has a correction at momentum $p \sim \hbar_{\text{eff}}/\xi_c$:

$$E^2 = p^2 c^2 (1 + \xi_{LIV} p^2 / M_{PT}^2) \quad (32)$$

$$M_{PT} = K_0^{1/2} / (\alpha^{1/2} \xi_c) \quad (33)$$

Required facility: Gamma-ray burst time delays with $E > 10$ TeV detectable by CTA or LHAASO.

Prediction 8: No-Go for Causal Loops (Structural Theorem)

The No-Closed-Phase-Loop Theorem (Section XI) is structural. Any confirmed observation of a closed timelike curve, retrocausal signal, or laboratory time-travel effect falsifies Phase Theory as a matter of logical necessity. Current bound: no violation observed at precision set by NIST clock networks.

SECTION X: LATTICE PROTOCOL AND THE KILLABLE NUMBER

X.1 Lattice Setup

- **Complex size:** 64^3 sites minimum (128^3 for continuum extrapolation).
- **Dimensions:** 2+1 for initial convergence checks; 3+1 for the physical run.

- **Action:** I_0 of Eq. 5 with uniform $K_e = K_0$ (isotropic), $\lambda_p/K_0 = 0.1$ (initial scan).
- **Update rule:** Gradient flow (Eq. 6) with adaptive step size; monitor I_0 per sweep.
- **Convergence diagnostic:** $\|g^{(n+1)} - g^{(n)}\|_{H^1}$ vs. n (the Lipschitz curve, Eq. 9).

X.2 The Killable Number: m_μ/m_e

Protocol:

1. **Step 1.** Seed two defect cores on the same lattice:
 - Electron defect: $\Xi_e = (k=-1, Q_H=0, t=1, \chi=-1/2, R=\text{singlet})$
 - Muon defect: $\Xi_\mu = (k=-1, Q_H=0, t=2, \chi=-1/2, R=\text{singlet})$
 - Same K_0 and λ_p for both.
2. **Step 2.** Minimize I_0 within each sector separately using conjugate gradient on $t^\vee$.
Record $I_{\min}(\Xi_e)$ and $I_{\min}(\Xi_\mu)$.
3. **Step 3.** Compute:

$$m_\mu / m_e|_{PT} = I_{\min}(\Xi_\mu) / I_{\min}(\Xi_e) \quad (34)$$

4. **Step 4.** Do **NOT** fit K_0 , λ_p , or any other parameter to the observed ratio 206.768.
Report the raw ratio with its statistical uncertainty from Monte Carlo variation of the minimization initial condition, and its systematic uncertainty from finite-lattice extrapolation (compare 64^3 vs. 128^3).
5. **Step 5. Verdict:**
 - *Within 20% of 206.768* at any point in the $K_0 - \lambda_p/K_0$ parameter space:
Phase Theory passes its first quantitative test.
 - *Incompatible with 206.768* across the entire accessible parameter space:
the radial-node mass generation mechanism is ruled out.

X.3 Expected Convergence Plot

The plot $\|\mathcal{G}^{(n+1)} - \mathcal{G}^{(n)}\|_{H^1}$ vs. n should exhibit three phases:

- **Phase 1** ($n = 0-5$): Rapid initial drop — adjustment from Minkowski seed.
- **Phase 2** ($n = 5-30$): Geometric decay — slope = $-\log(1/L)$ per step, $L = \alpha C_K$ the measured Lipschitz constant.
- **Phase 3** ($n > 30$): Plateau — numerical noise floor.

Required Artifact

This convergence plot is the primary artifact required before any phenomenology claim is credible. Without it, Theorem 1 remains a theoretical claim. With it, Phase Theory has its first piece of numerical evidence that any physicist can independently verify.

SECTION XI: CAUSALITY AND THE NO-CLOSED-PHASE-LOOP THEOREM

XI.1 The Update Parameter

The update parameter u in Eq. 6 is not physical time. Physical time t is a derived quantity constructed from the partial ordering $<$ of the DAG D (Definition 3) via calibration against the emergent metric. The mapping $u \rightarrow t$ is monotone but non-linear; a denser, more rapidly updating region of the DAG corresponds to a faster proper time.

XI.2 Theorem 2 (No-Closed-Phase-Loop)

Theorem 2 (No-Closed-Phase-Loop)

In any physical phase configuration consistent with Definitions 1–6, no closed causal loop exists.

Proof: By Definition 3, D is a DAG; its transitive closure $<$ is a strict partial order (irreflexive, asymmetric, transitive). Suppose $e_1 < e_1$. Irreflexivity gives a contradiction. Suppose $e_1 < e_2 < e_1$. Transitivity gives $e_1 < e_1$, contradicting irreflexivity. By induction on n , any cycle $e_1 < e_2 < \dots < e_n < e_1$ gives $e_1 < e_1$, a contradiction. Therefore no closed causal loop exists.



Corollary 3: Time travel, closed timelike curves, and all causal paradoxes are structurally impossible in Phase Theory — as theorems, not additional postulates.

SECTION XII: QUANTUM MECHANICS AS AN EFFECTIVE DESCRIPTION

Small fluctuations $\delta\varphi$ about the minimizer Φ_* satisfy the linearized equation (Eq. 15). Restricted to a finite coherence domain, the space of such fluctuations is a finite-dimensional Hilbert space H with inner product:

$$\langle \delta\varphi_1, \delta\varphi_2 \rangle = \delta^2 I_0[\delta\varphi_1, \delta\varphi_2] \quad (35)$$

This is positive-definite by the stability of Φ_* . The Schrödinger equation emerges from the gradient flow (Eq. 6) restricted to the linear fluctuation sector:

$$i \hbar_{\text{eff}} d/dt |\psi\rangle = \hat{H} |\psi\rangle, \quad \hat{H} = \delta^2 I_0 / \Phi_* \quad (36)$$

The Born rule follows from the quadratic nature of I_0 : the relative weight of a sector $[\Xi]$ in a superposition is $\|\delta\varphi_{[\Xi]}\|^2 / \|\delta\varphi\|^2$, which is exactly the Born rule.

SECTION XIII: ANTIMATTER, CONFINEMENT, AND GAUGE STRUCTURE

XIII.1 Antimatter

The antiparticle of $\Xi = (k, Q_H, t, \chi, R)$ is $\bar{\Xi} = (-k, -Q_H, t, -\chi, \bar{R})$. Mass equality $m(\Xi) = m(\bar{\Xi})$ follows because I_0 depends on $|D_e \Phi|^2$ (a norm), which is orientation-independent. CPT invariance is a topological identity.

XIII.2 Confinement

A single quark carries holonomy R in the fundamental representation of $\text{SU}(3)_c$. For finite I_0 , the holonomy over any contractible cycle must reduce to the identity. A single quark's holonomy is a non-trivial $\text{SU}(3)$ element on any enclosing 2-sphere; this gives $I_0 = \infty$. Isolated quarks are topologically forbidden. The linear potential $V(r) = \sigma r$ is the energy cost of the flux tube between separated quarks, with string tension σ computable from K_e in the $\text{SU}(3)$ sector.

XIII.3 Maxwell and Yang-Mills

Linearizing I_0 around the vacuum in the $\text{U}(1)_Y$ sector:

$$I_0^{\text{lin}}[A] = (1/4) \int F_{\mu\nu} F^{\mu\nu} d\mu_{g^*} \quad (37)$$

Euler-Lagrange equations: $\partial_\mu F^{\mu\nu} = J^\nu$ (Maxwell). Non-Abelian sectors: $D_\mu F^{\mu\nu} = J^\nu$ (Yang-Mills). Both emerge from the same functional I_0 ; no separate gauge action is introduced.

SECTION XIV: BLACK HOLES, INFORMATION, AND ENTROPY

A black hole is a region where $\langle |D\Phi|^2 \rangle \rightarrow C_K$ (coherence saturation). The emergent metric (Theorem 1) is no longer a contraction at this point — the metric description breaks down. The field is regular; the effective geometry is undefined. The information is encoded in the topological state of the boundary (the horizon), consistent with the area scaling:

$$S = A / (4\ell_P^2) \quad (38)$$

which is a consequence of the counting measure on the simplicial boundary complex. Hawking radiation is a tunneling process in the boundary simplicial complex, with the modified spectrum given by Eq. 30.

SECTION XV: COSMOLOGY AND THE DARK SECTOR

XV.1 The Big Bang

The Big Bang is a phase-ordering transition from a maximally disordered initial state ($I_0 \rightarrow \infty$ per site, all holonomies maximally non-trivial). The “singularity” is the breakdown of the continuum approximation; the discrete simplicial complex remains well-defined. Inflation is the coherent ordering phase, generating a nearly scale-invariant (Harrison-Zel’dovich) fluctuation spectrum.

XV.2 Dark Matter

Dark matter corresponds to topological sectors with $k = 0$ (no $U(1)_Y$ winding) and R a singlet of $SU(3)_c$ (no color) but $Q_H \neq 0$ (non-zero Hopf charge — gravitationally active through I_0 mass assignment). Direct-detection recoil spectrum is given by Eq. 25.

XV.3 Dark Energy

The cosmological constant Λ in Eq. 20 is the background phase potential $\mathcal{V}(\Phi_*)$ at the vacuum — the minimum of I_0 in the vacuum sector. This is a finite, calculable number, not a divergent zero-point energy sum. The cosmological constant problem is reframed as a well-posed discrete minimization. The numerical value of $\mathcal{V}(\Phi_*)$ is a lattice measurement target.

SECTION XVI: OPEN PROBLEMS

Table 4. Open Problems Requiring Resolution

Problem	Status	Path to Resolution
Derivation of 3+1 dimensionality from stability analysis	Partial argument (Section II.1)	Rigorous spectral analysis of defect stability on C_n complexes
Numerical value of G_{eff} from first principles	Open; G_{eff} defined in Eq. 19 but $\langle D\Phi ^2 \rangle_*$ requires lattice	Lattice measurement of $\langle D\Phi ^2 \rangle_*$ at fixed K_0, α
Stability of $t = 4$ mode (fourth generation)	Upper-bounded by energy argument; not numerically confirmed	Lattice minimization in $t = 4$ sector
Full non-perturbative anomaly cancellation	$\mathbb{Z}_6$ quotient argument correct but requires complete index theorem check	Atiyah–Singer index computation on C_0

Problem	Status	Path to Resolution
Quantitative inflation model (reheating)	Not yet specified	Connect phase-ordering dynamics to CMB C_ℓ spectrum

SECTION XVII: COMPARISON WITH EXISTING FRAMEWORKS

Table 5. Framework Comparison

Framework	Metric	Gauge Group	Particle Spectrum	Falsifiers
Standard Model + GR	External input	Postulated	Postulated (19 parameters)	High (but parameters are free)
String/M-theory	External (mostly)	Derived from Calabi-Yau	Landscape ($\sim 10^{500}$)	Contested
Loop Quantum Gravity	Emergent	Not unified with SM	Not specified	Low at current energies
Causal Set Theory	Emergent	Not unified	Not specified	Very low
Phase Theory	Fixed-point emergent (Theorem 1)	Derived from π_1 , π_3 , anomaly (Prop. 1)	Finite, topology-indexed	8 classes, metrological

SECTION XVIII: CONCLUSION

Phase Theory is a logically closed pre-metric framework in which:

6. The circular metric bootstrap is resolved by Theorem 1: the metric g^* is the unique fixed point of a Banach contraction on $\text{Sym}^2 T^*M$, initialized at Minkowski and iterating a metric-independent discrete averaging. The convergence rate is the measurable Lipschitz constant $L = \alpha C_K$.
7. The Standard Model gauge group is derived, not assumed: it is the unique minimal compact Lie group satisfying $\pi_1(T) = \mathbb{Z}$, $\pi_3(T) = \mathbb{Z} \oplus \mathbb{Z}$, and anomaly freedom (Proposition 1). The three-generation structure follows from the radial node count of the Hopf map.
8. Klein–Gordon, Dirac, and linearized Einstein–Hilbert gravity emerge from a single quadratic expansion of I_0 at the fixed point Φ^* . The constants $\hbar$, c , G are expressed analytically in Eqs. 16 and 19; only G requires one lattice measurement ($\langle |D\Phi|^2 \rangle_*$).
9. Eight falsifiable predictions are stated with metrological specificity: central value forms, error budget sources, required exposures, and falsification criteria are given for LIGO O4/O5, XENONnT/LZ, ALPHA-g/GBAR, and CMB-S4.
10. The killable number is m_μ/m_e (Eq. 34): computable from two lattice minimizations at identical K_{ab} , without fitting the result to the observed ratio 206.768.

The next required step before journal submission is the lattice convergence artifact: run the 64^3 discrete action, plot $\|g^{(n+1)} - g^{(n)}\|_H^1$ vs. n , and report the measured Lipschitz constant L . Without this plot, Theorem 1 remains a claim about a hypothetical lattice. With it, Phase Theory has its first piece of numerical evidence that any physicist can independently verify.

APPENDIX A: NOTATION GLOSSARY

Table A1. Symbol Definitions

Symbol	Definition
$C_0 = (V, E, P, \sigma)$	Finite oriented simplicial complex (base ontology)
μ_0	Counting measure on V ; $\mu_0(A) = A $
$\Phi : V \rightarrow T$	Phase field; sole dynamical variable
$D_e \Phi$	Edge derivative = $\log(\Phi(i)^{-1} \Phi(j)) \in \mathfrak{t}$
$\text{hol}_p(\Phi)$	Holonomy around plaquette p
$I_0[\Phi]$	Discrete inconsistency functional (Eq. 5); metric-free
$g^{(n)}$	Metric at iteration step n ; $g^{(0)} = \eta$ (Minkowski)
g^*	Fixed-point metric (Theorem 1)
$L = \alpha C_K$	Lipschitz constant of metric iteration; measurable from lattice
$T_{\min}$	Derived target manifold = $(U(1)_Y \times SU(2)_L \times SU(3)_c)/\mathbb{Z}_6$
$\Xi = (k, Q_H, t, \chi, R)$	Particle invariant quintuple
$\varepsilon = (\xi_d/L)^2$	GR correction parameter; $\varepsilon \sim 10^{-3}$ at ringdown
$\hbar_{\text{eff}}$	$(K_0 \alpha)^{1/2}$
G_{eff}	$[8\pi\alpha K_0 \langle D\Phi ^2 \rangle_*]^{-1}$
$f_{\text{NL}}^{\text{top}}$	Topological non-Gaussianity parameter (CMB)
Λ_{top}	Topological momentum scale = $\hbar_{\text{eff}}/\xi_c$

APPENDIX B: PROOF SKETCHES

Theorem 1 (Full)

The map $F : g \rightarrow g'$ defined by Eq. 7 satisfies $\|Fg - F\tilde{g}\|_{L^\infty} \leq \alpha C_K \|g - \tilde{g}\|_{L^\infty}$. Under H3 ($\alpha C_K < 1$), Banach's theorem gives a unique fixed point g_ with exponential convergence. The fixed point lies in the interior of $B(L^\infty$ ball of radius $1/2$ around η) by smallness of αC_K .*

Proposition 1 (Full)

Anomaly cancellation for a single generation of defects with quantum numbers (k, Q_H, R) under $U(1)_Y \times SU(2)_L \times SU(3)_c$ requires five conditions: $\Sigma Q_Y^3 = 0$, $\Sigma Q_Y = 0$, $[SU(2)]^2 U(1) = 0$, $[SU(3)]^2 U(1) = 0$, and mixed gravitational anomaly cancellation. With quarks in the $SU(3)$ fundamental (3 colors) and leptons as $SU(3)$ singlets, these are satisfiable if and only if the hypercharge assignments are $Q_Y(u_R) = 4/3$, $Q_Y(d_R) = -2/3$, $Q_Y(e_R) = -2$, $Q_Y(L) = -1$, $Q_Y(Q) = 1/3$ — precisely the Standard Model assignments. The $\mathbb{Z}_6$ quotient is the unique discrete identification making all representations well-defined on T_{min} .

Theorem 2 (Full)

Proved in Section XI.2 by irreflexivity and transitivity of the strict partial order on the DAG D .

APPENDIX C: STANDARD MODEL MAPPING TABLE

Table C1. Particle Invariant Assignments for All Standard Model Particles

Particle	$\Xi = (k, Q_H, t, \chi, R)$	Mass Source
Electron (e^-)	$(-1, 0, 1, -1/2, \text{singlet})$	$I_{\min}$ at $t=1$
Muon (μ^-)	$(-1, 0, 2, -1/2, \text{singlet})$	$I_{\min}$ at $t=2$
Tau (τ^-)	$(-1, 0, 3, -1/2, \text{singlet})$	$I_{\min}$ at $t=3$
Electron neutrino (ν_e)	$(0, 0, 1, +1/2, \text{singlet})$	Near-zero; distinct topological sector
Up quark (u)	$(+2/3, (1,0), 1, \pm 1/2, \mathbf{3})$	$I_{\min}$, SU(3) fundamental
Down quark (d)	$(-1/3, (1,0), 1, \pm 1/2, \mathbf{3})$	$I_{\min}$, SU(3) fundamental
Photon (γ)	$(0, 0, 0, \pm 1, \text{singlet})$	Propagating U(1) mode; $m=0$
$W^\pm$ bosons	$(\pm 1, (1,0), 0, \pm 1, \mathbf{2})$	SU(2) propagating mode
Z^0 boson	$(0, (1,0), 0, \pm 1, \text{singlet})$	Mixed U(1) $\times$ SU(2) propagating mode
Gluon (g)	$(0, (0,1), 0, \pm 1, \mathbf{8})$	SU(3) adjoint propagating mode
Higgs (H)	$(0, 0, 0, 0, \text{singlet})$	Radial excitation of $\mathcal{V}(\Phi)$

REFERENCES

- [1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (Westview Press, 1995).

- [2] S. Weinberg, *The Quantum Theory of Fields*, Vols. 1–3 (Cambridge University Press, 1995–2000).
- [3] G. 't Hooft and M. Veltman, Ann. Inst. Henri Poincaré **20**, 69 (1974).
- [4] M. H. Goroff and A. Sagnotti, Nucl. Phys. B **266**, 709 (1986).
- [5] J. Polchinski, *String Theory*, Vols. 1 & 2 (Cambridge University Press, 1998).
- [6] M. B. Green, J. H. Schwarz, and E. Witten, *Superstring Theory* (Cambridge University Press, 1987).
- [7] C. Rovelli, *Quantum Gravity* (Cambridge University Press, 2004).
- [8] T. Thiemann, *Modern Canonical Quantum General Relativity* (Cambridge University Press, 2007).
- [9] J. Ambjorn, J. Jurkiewicz, and R. Loll, Phys. Rev. Lett. **95**, 171301 (2005).
- [10] R. Loll, Class. Quantum Grav. **37**, 013002 (2020).
- [11] R. D. Sorkin, in *Lectures on Quantum Gravity* (Springer, 2005).
- [12] F. Dowker, in *100 Years of Relativity* (World Scientific, 2005).
- [13] T. Jacobson, Phys. Rev. Lett. **75**, 1260 (1995).
- [14] J. D. Bekenstein, Phys. Rev. D **7**, 2333 (1973).
- [15] S. W. Hawking, Commun. Math. Phys. **43**, 199 (1975).
- [16] ALPHA Collaboration, Nature **621**, 716 (2023).
- [17] E. Aprile et al. (XENON Collaboration), Phys. Rev. Lett. **131**, 041003 (2023).
- [18] K. N. Abazajian et al. (CMB-S4 Collaboration), arXiv:1610.02743 (2016).
- [19] Planck Collaboration, A&A **641**, A6 (2020).
- [20] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), Phys. Rev. Lett. **116**, 061102 (2016).
- [21] J. Steinhauer, Nature Physics **12**, 959 (2016).
- [22] M. Hanks, Phase Theory v1–v2, Preprint (2025–2026).

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